

Metrics for Scientific Discovery: Auditing Embedding Spaces for Novelty, Calibration, and Interpretability

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Abstract

Self-supervised objectives with explicit distributional priors — LeJEPa and its sketched-isotropic-Gaussian regularizer (SIGReg) being the canonical example — promise embedding spaces whose *geometry*, not merely whose linear separability, carries information. This promise matters most in scientific settings, where the object of interest is by construction *unlabeled*: a new particle, a new galaxy morphology, a new material phase. Yet the field evaluates such spaces almost exclusively with linear probes, which measure exactly the one property a discovery pipeline cannot rely on. We argue that a space intended for discovery must be audited in four distinct currencies — *detection*, *calibration*, *discoverability*, and *interpretability* — and we supply a concrete, reproducible battery for each. Using a campaign of 79 controlled experiments spanning CIFAR-10/100 and a $4 \times 3 \times 6$ grid of transfer datasets (Stanford Cars, Flowers-102, DTD, Galaxy10 DECals) $\times$ pretraining bases (DINO, LeJEPa, VISReg) $\times$ contrastive/regularized objectives, we establish four results. (i) Detection and calibration are *empirically dissociated*: softmax-contrastive spaces win the probe and score exactly 0.000 per-event detection power, while density-ratio-calibrated spaces invert the ordering; no CIFAR-100 space in our sweep exceeds probe 0.94 with tied-Mahalanobis AUC > 0.42 . (ii) The dissociation is *repairable by construction rather than by loss design*: residual fine-tuning, which trains a second head on $\mathbf{z} - \boldsymbol{\mu}_y$ and concatenates, beats the alternative (iterated anchor discovery) in 12/12 dataset $\times$ base cells and yields the first spaces that hold a record probe and calibrated distances simultaneously. (iii) *Scale-free* statistics see novelty that distance-based scores cannot: local intrinsic dimension (LID) reaches AUC = 0.951–0.955 on Flowers-102 using *no novelty labels*, exceeding the best supervised probe (0.906) — the only unsupervised score to beat a trained probe anywhere in the campaign. (iv) Interpretability is a *measurable, dissociable* property: superclass agreement and dendrogram purity track supervision, not probe quality, and a never-labeled held-out class reliably lands among its true semantic siblings. We distill the results into a reporting protocol and a set of regime rules — which objective, which statistic, which repair — for practitioners building embedding spaces meant to find things nobody has labeled.

1 Introduction

Representation learning is evaluated, overwhelmingly, by how well a linear classifier reads a representation out. This is a defensible proxy when the downstream task is classification into known categories. It is the wrong proxy when the downstream task is *discovery*.

A discovery pipeline in the physical sciences — searching for a resonance in collider data, an anomalous transient in a survey, an unfamiliar defect in a microscopy campaign — asks a different question of an embedding. It asks: *given a reference population, is this event unlike anything I have seen, and by how much?* Answering requires the space to support an absolute notion of distance, so that a threshold set on simulated background transfers to data. It requires the tail of the distance distribution to be enriched in genuine novelty, so that clustering the tail recovers new categories

rather than mislabeled old ones. And — because a discovery must eventually be *explained* — it requires the geometry to be semantically legible, so a domain expert can say what the new cluster sits next to and why.

None of these are linear separability. All of them are measurable.

The LeJEPA opening The recent line of work that regularizes embeddings toward an explicit distributional target — most directly LeJEPA and its sketched isotropic Gaussian regularizer (SIGReg), which penalizes the sliced 1-D Wasserstein-2 distance between the embedding and $\mathcal{N}(0, I)$ — changes what is available to measure. Once a marginal is *specified* rather than emergent, distances have units: a Mahalanobis radius means something, a density ratio can be calibrated, and a likelihood-based anomaly score becomes well posed. Our starting hypothesis is that this is the correct substrate for discovery, and that its evaluation should exploit the structure it provides instead of discarding it in favor of a probe.

That hypothesis turns out to be half right, in an informative way. A Gaussian marginal is necessary for calibrated geometry but not sufficient for detection, and the two properties trade off sharply enough that we treat them as separate axes throughout.

Contributions

- **A four-currency evaluation battery** (§3) for embedding spaces intended for discovery: detection, calibration, discoverability, interpretability — with concrete estimators, standard protocols, blind spots, and known failure modes for each.
- **Systematic evidence that the currencies dissociate** (§4), across 79 experiments, two image regimes, four transfer datasets, three pretraining bases, and ten training objectives. The dissociation is not noise and is not a tuning artifact: it is reproduced under paired seeds and inverts predictably with the number of seen classes.
- **Two repairs, benchmarked head to head** (§5): iterated anchor *discovery* and *residual fine-tuning*. Residual fine-tuning wins 12/12 cells and produces the campaign’s records; discovery is shown to be *substrate-dependent* (probe-negative on frozen trunks, probe-positive in 50/72 cells on fine-tuned trunks) and *purity-gated*.
- **Local intrinsic dimension as a first-class novelty statistic** (§6), including a full control study: holdout rotation, distance baselines, bootstrap intervals, per-class decomposition, and k -sensitivity. It sets a new unsupervised record on Flowers-102 and, importantly, we characterize where it *fails*.
- **Interpretability as a metric, not a figure** (§7): superclass agreement@ k , dendrogram purity, centroid silhouette, and cross-space similarity (CKA, mutual- k NN, Procrustes, ridge R^2 , TwoNN intrinsic dimension), applied to every space in the campaign.

2 Setup

Open-world protocol In every experiment a set of classes is held out *entirely*: their images and their labels are removed from the training corpus, so at evaluation time they are genuinely novel. Unless noted, the held-out set is the final ten classes of the dataset (Galaxy10: the final one; CIFAR: class 4). Embeddings are then frozen and scored. This is stricter than the common “train-on-everything, test-novelty-by-relabeling” protocol, and we flag the one legacy table (§4.1, aircraft trunk reuse) where an earlier run violated it.

Objectives All objectives are instances of one design cube, which lets us attribute effects to axes rather than to named methods. An objective is specified by four coordinates:

$$(\text{positives}) \times (\text{critic}) \times (\text{estimator}) \times (\text{marginal}).$$

Positives are instance (augmentation) pairs or supervised (same-label) pairs. The *critic* g is cosine $\langle \hat{\mathbf{z}}, \hat{\mathbf{z}}' \rangle / \tau$, bilinear $\langle \mathbf{z}, \mathbf{z}' \rangle / \tau$, or distance $-\frac{1}{2} \|\mathbf{z} - \mathbf{z}'\|^2 / \tau$. The *estimator* is either the standard row-normalized softmax (recovering NT-Xent/SupCon) or the un-normalized maximum-likelihood form

$$\mathcal{L}_{\text{NPLM}} = \mathbb{E}_{\text{ref}}[e^g - 1] - \mathbb{E}_{\text{pos}}[g], \quad (1)$$

dataset	classes	held out	seen train pool	character
CIFAR-10	10	{4}	45 000	coarse, many samples/class
CIFAR-100	100	{4}	49 500	many-class, low resolution
Stanford Cars	196	186–195	7 712	fine-grained (car models)
Flowers-102	102	92–101	1 840	fine-grained, few-shot (~ 18 /class)
DTD	47	37–46	2 960	texture
Galaxy10 DECals	10	{9}	1 585	coarse morphology, 10%/90% split
FGVC-Aircraft	100	90–99	$\sim 6\,600$	fine-grained (variants)

Table 1: Datasets and open-world splits. Transfer bases are DINO, LeJepa and ViT-B/16 trunks (768-D CLS features); CIFAR uses a ResNet-20 trunk (CIFAR-pretrained, or randomly initialized for the from-scratch lineages).

arm	positives	critic	estimator	marginal	labels
simclr	instance	cosine	softmax	—	no
lejepa	MSE invariance + $\lambda \cdot \text{SIGReg}$			global	no
ss (simclr_sigreg)	instance	cosine	softmax	global	no
supcon	supervised	cosine	softmax	—	yes
supcon_sigreg	supervised	cosine	softmax	global	yes
nplm_bilinear	instance	bilinear	NPLM	global	no
nplm_distance	instance	distance	NPLM	global	no
nplm_sup_dist	supervised	distance	NPLM	global	yes
nplm*_cw	(either)	(either)	NPLM	classwise	varies
sup (recipe)	prototype CE to anchors + repulsion			classwise	yes

Table 2: The objectives as coordinates in the design cube. lejepa and the sup recipe are not contrastive and are listed by their components. Standard training: 20 epochs, Adam 10^{-3} , batch 256 two-view (balanced 24/class for classwise arms), $\lambda = 1$.

whose minimizer is the *absolute* log density ratio $g^* = \log \frac{p(\mathbf{x}, \mathbf{x}')}{p(\mathbf{x})p(\mathbf{x}')}$ with the calibration constraint $\mathbb{E}_{\text{ref}}[e^{g^*}] = 1$. The *marginal* is none, global SIGReg, or classwise SIGReg. The distinction in the estimator axis is the crux of this paper: softmax self-normalizes per row and therefore recovers pointwise mutual information only up to an unknown per-row constant, destroying absolute scale; Eq. 1 does not. We verify the calibration empirically against analytic PMI on a controlled toy ($r > 0.9$).

Two knobs, one inert Two hyperparameters govern the NPLM arms, and they are not symmetric. At initialization the interaction term of Eq. 1 is $\sim 10^{10}$ while $\lambda \cdot \text{SIGReg}$ is $\sim 10^{-2}$: λ is *inert for the first ~ 2 epochs* and only shapes the endgame. By contrast τ is a sharp optimum ($\tau = 1$ for bilinear on CIFAR-100; $\tau \in \{0.5, 2\}$ each cost 0.06–0.10 probe AUC). We therefore scan τ and report λ ablations only where the marginal type changes.

3 A four-currency battery for discovery

We propose that any embedding space advertised for discovery be reported in four currencies. The first two are per-space properties; the third is a property of a *pipeline*; the fourth is a property of the space’s relation to human categories.

3.1 Currency I — Detection

Can novelty be read out at all? Estimator: the AUC of a linear holdout-vs-rest probe trained on frozen embeddings, reported as mean $\pm$ sd over 3 seeds. This is the standard metric and we retain it, with two caveats that recur throughout: single-seed spread is ≈ 0.017 , so **gaps below 0.02 require 3–5 seeds**; and the probe is blind to radial and non-linear novelty structure, which is precisely the structure the other currencies measure.

We also report nearest-centroid top-1 accuracy over seen classes (acc) and the macro one-vs-rest AUC of the prototype posterior (supAUC) as closed-set sanity checks — a space can win the novelty probe while being useless at its nominal task, and we have observed exactly that (nplm-bil-ft on Flowers: probe 0.808 post-discovery, acc 0.531).

3.2 Currency II — Calibration

Do distances have absolute meaning? A discovery threshold is set on a reference population and applied to data; if the score’s scale drifts, the threshold is worthless even when the space is geometrically fine. We estimate calibration at two levels.

Per-space: novelty AUC of the minimum distance to a seen-class centroid (euc1), and of tied and per-class Mahalanobis distances (mahaT, mahaPC; shrinkage 0.1). Plus a Gaussianity summary — eigenvalue conditioning, class RMS, maximum off-diagonal correlation, sliced-Wasserstein ratio, skew and kurtosis — which audits whether the SIGReg target was actually reached rather than merely optimized.

Per-event: the fraction of injected signal events exceeding the $\alpha = 0.05$ null quantile of the reference score distribution. This is the metric that most sharply separates the objective families, and it is the one closest to how a physics search actually operates. **SupCon-family spaces score 0.000 here**, on every dataset, at every signal fraction: a self-normalized objective simply does not produce a transferable threshold.

Dataset-level: three two-sample batteries — SparKer (semi-parametric kernel discrepancy, $M=16$ kernels), Mahalanobis, and MMD — run as toy experiments (N_D events per toy, signal fraction f injected from the holdout pool, power = fraction of 50 signal toys above the $\alpha = 0.05$ quantile of 200 null toys). Two lessons are protocol-critical. First, **kernel statistics must use an annealed median-heuristic bandwidth**: a fixed $\sigma=1$ is calibrated only for the unit-class-width supervised recipe, and a discovery fine-tune that rescales a compact trunk by $\sim 4\times$ sends fixed- σ SparKer to zero power at every fraction *while the space is fine* (annealed: 1.00). We regard this as a general warning about kernel two-sample tests across heterogeneous geometries. Second, statistic and geometry must be matched: parametric statistics need calibrated spaces, and Mahalanobis is simply dead on CIFAR-100 at every dimension and every space (inter-class similarity), where MMD remains the dimension-robust choice.

3.3 Currency III — Discoverability

Does the tail contain the new thing? A discovery loop pools events past a high quantile (0.95) of the seen-distance distribution, selects a cluster count by BIC, merges nearby anchors, and fine-tunes the space on seen labels plus the resulting pseudo-labels. The diagnostic that matters is **pool purity**: the fraction of the pooled tail that is genuinely novel. Everything else about the pipeline is downstream of it.

Purity varies over two orders of magnitude across our datasets — 0.795 on DTD, 0.33–0.66 on Flowers, 0.34 on CIFAR-10, 0.03–0.14 on Cars, 0.003–0.013 on CIFAR-100 — and the variation is *structural*, not a tuning failure. On CIFAR-100, 500 holdout images sit inside a ~ 2500 -event tail: the loop is rate-blocked before it begins. On Cars, purity caps at 0.14 for a deeper reason we consider one of the campaign’s cleanest negative results: **a novel car model looks like existing car models**. Pooling requires novelty that is geometrically *outlying*, and fine-grained novelty is not, calibrated space or no. We recommend reporting purity alongside any claimed discovery gain, because a gain at low purity is mechanistically something else (§5).

3.4 Currency IV — Interpretability

Is the geometry legible? Where a superclass partition exists (CIFAR-100 coarse labels, aircraft manufacturer, car make, galaxy morphology) we compute: nearest-class-centroid superclass agreement@1 and @5 against chance, average-linkage dendrogram purity, centroid silhouette, and the within/between distance ratio. Where it does not, we report top-5 nearest-centroid tables, which are directly auditable by a domain expert.

We additionally characterize how spaces relate to *each other*, on aligned samples ($n=4000/\text{cell}$): linear CKA, mutual $k\text{NN}@10$, LLE-weight transfer, orthogonal Procrustes, directional ridge-map

space (CIFAR-10, 32-D, holdout 4)	probe	acc	eucl	mahaT	per-event
supcon	0.942	0.861	0.689	0.644	0.000
supcon_sigreg	0.882	0.844	0.637	0.577	0.000
sup (supervised SIGReg)	0.923	0.884	0.778	0.645	—
nplm_sup_dist	0.783	0.871	0.726	0.723	0.20
nplm_bilinear (no labels)	0.853	0.546	0.510	0.494	—
simclr	0.867	0.729	0.236	0.229	0.000
lejepa	0.864	0.695	0.408	0.371	—

Table 3: The dissociation in one table. The probe ordering and the calibration ordering are close to reversed. Note `simclr`’s `eucl` of 0.236 — *below* chance-adjusted expectations for a space with a probe of 0.867: the information is there and is not in the distances.

R^2 , calibration transfer through a fitted ridge map, and TwoNN intrinsic dimension. This turns “these two objectives learn different things” from an assertion into a measurement, and it is how we establish that the NPLM spaces are *reorganizations* of information the softmax spaces already contain rather than new information (§7).

4 The central dissociation

The single most robust finding across the campaign is that Currency I and Currency II are *anti-correlated across spaces*. Softmax-contrastive objectives maximize linearly readable structure and carry no absolute distance information; density-ratio-calibrated objectives are ready-to-use metric spaces at a probe cost.

On CIFAR-100 the effect is sharp enough to state as a constraint: **no space in the sweep exceeds probe 0.94 with mahaT > 0.42**, across 30+ standalone spaces at 32/64/100-D, hub-initialized and from-scratch. The probe ceiling (≈ 0.95) is owned by supervised softmax and prototype recipes; the mahaT ceiling (≈ 0.47 – 0.49) is owned by NPLM arms. The frontier between them is populated only by *concatenations* (§5).

This is not a seed artifact The one place we initially believed the dissociation was breakable — `nplm.bilinear` reaching probe 0.935 on CIFAR-100, apparent parity with `supcon` — dissolved under paired multi-seed testing: over 5 paired seeds `supcon` gives 0.940 ± 0.004 against `nplm.bilinear`’s 0.855 ± 0.042 , paired difference -0.085 ($t = -4.7$). The single-seed 0.935 was a tail draw from a heavy-tailed seed distribution. The $10\times$ larger seed variance of the label-free NPLM arm is itself an unexplained and, we think, interesting phenomenon.

The dissociation recurs at every level It is not a property of the loss family alone. It reappears (i) in the fine-tuning objective: prototype/repulsion fine-tuning wins the probe (0.97 vs 0.81–0.84) while NPLM+SIGReg fine-tuning wins *every* power statistic (per-event 0.57 vs 0.30–0.48; SparKer 1.00/0.92; Mahalanobis 0.90; MMD 0.74) and holds round-2 pool purity (0.22–0.30 vs 0.03–0.11); (ii) in the residual construction (§5), where NPLM residuals preserve calibration and plain residuals buy probe with it; and (iii) in semantic structure (§7), where the calibration-strong label-free arms are also the semantically emptiest. The practical rule is uncomfortable but clear: **choose by consumer**. Linear readout $\rightarrow$ prototype/softmax. Detection with a transferable threshold $\rightarrow$ NPLM.

4.1 Regime rules: which corner wins where

The winning objective is determined by regime, and — importantly for practitioners — is *base-invariant*: the same arm wins on DINO, LeJEPa and ViSReg trunks. The governing variable is the number of seen classes relative to the embedding dimension, modulated by samples/class.

Three sub-rules are worth isolating.

Anchor realizability gates the classwise marginal. Classwise SIGReg is strictly better than global SIGReg *iff* the class anchors are reachable, i.e. $\dim \geq n_{\text{classes}}$. With 100 anchors in 32-D or 64-D, centroids stall ≈ 3.5 away from their targets and the classwise marginal becomes actively harmful

regime	example	probe champion	calibration champion
few classes, coarse	CIFAR-10, Galaxy10	supcon	nplm_sup_dist
many classes, fine-grained	Cars, Flowers, Aircraft	ss/supcon_sigreg	ss
texture	DTD	simclr	ss
many classes, low-res	CIFAR-100	supcon	nplm_bilinear

Table 4: Regime rules, replicated on all three pretraining bases. Note the inversions: the SIGReg marginal *helps* on fine-grained low-data transfer and *hurts* on CIFAR; label-free instance discrimination wins outright on texture, where labels add nothing to the probe.

for label-free arms. This is a geometric precondition, checkable before training, and we recommend reporting the centroid-to-anchor distance as a routine diagnostic.

High dimension is not free for calibrated objectives. Moving CIFAR-100 from 32-D to 64-D causes the bilinear NPLM arms to *regress on both currencies* ($0.935 \rightarrow 0.858$; $0.925 \rightarrow 0.825$), while softmax and supervised distance arms gain mildly. On Cars at 100-D the same signature appears from the other side: nplm_sup_dist gains the most probe ($+0.096$) but loses its per-event calibration lead ($0.106 \rightarrow 0.03$). At high class counts the extra dimensions are spent on probe-readable directions, not calibrated ones.

Fine-tuning inverts the base ranking. On frozen features DINO dominates every currency. After end-to-end fine-tuning the ordering largely dissolves — VISReg holds several pre-discovery records (DTD simclr-ft 0.854, Cars ss-ft 0.741). Cross-space CKA supplies the mechanism (§7): supervision pulls every base toward one shared geometry.

Cross-entropy is not the ceiling. On every substrate tested, a supcon_sigreg head beats a supervised CE head on every column, including — on a LeJEPa fine-tuned trunk — beating CE’s own trained-classifier top-1 with mere nearest-centroid accuracy (0.638 vs 0.625).

5 Repairing the dissociation

Given that no single standalone loss in the design cube achieves both currencies, we benchmark two *constructions* that might.

5.1 Discovery is substrate-dependent and purity-gated

The iterated discovery loop (§3, Currency III) behaves in a way that is initially confusing and ultimately simple. On frozen trunks it is *probe-negative* on Cars for every arm (-0.02 to -0.09). On end-to-end fine-tuned trunks it is *probe-positive* for all six arms ($+0.03$ to $+0.11$) — *with pool purity still below 0.14 in both cases*. Across the full grid it is positive in 50/72 cells, and the supcon-ft+discovery pipeline is positive in 12/12 dataset×base cells, the only universally safe full pipeline we found.

Since purity did not change, the gain cannot be coming from pure pools. The mechanism is that impure-pool prototype fine-tuning acts as *beneficial refinement* when the substrate is already aligned to the task, and as *erosion* when it is not. This reframes discovery: on low-purity data it is not finding new classes, it is a regularizer, and it should be reported as such. Consistently, the largest single gain in the program is a *rescue* — nplm-sup-ft on DTD, $0.586 \rightarrow 0.781$, replicated at $+0.14$ to $+0.20$ on all three bases — recovering what a weak head missed rather than discovering anything.

Where purity *is* high, discovery does what it advertises: CIFAR-10 (purity 0.34) yields the best full pipeline on record, and Flowers (purity 0.33–0.66) yields $+0.106$ to $+0.145$. But purity is **necessary and not sufficient**: aircraft pools at moderate purity (0.32–0.53) and discovery still costs probe (-0.012 to -0.037) while buying geometry and per-event power. And Flowers/VISReg shows purity 0.65+ can leave a strong arm flat — there was nothing left to learn.

One pathology deserves flagging for anyone deploying such a loop: ss-ft discovery on the LeJEPa base collapses the probe $0.792 \rightarrow 0.613$, the worst outcome on record, by splitting round 1 into six anchors at purity 0.605 that fragment the *strongest* classes rather than the novelty. High purity does not protect against a bad partition.

dataset	best construction	probe (per-event)	parent	prior champion
Aircraft/VISReg	supcon→res-nplm cat	0.863 (0.363)	0.809	0.812 [†]
Cars/VISReg	supcon→res-nplm cat	0.855 (0.263)	0.707	0.759
Flowers/DINO	supcon→res-nplm cat	0.885 (0.334)	0.708	0.814
DTD/VISReg	supcon→res cat	0.847 (0.077)	0.750	0.854 [‡]
Galaxy10/LeJEPa	supcon→res cat	0.975 (0.001)	0.919	0.946
CIFAR-100	supcon→res-nplm cat	0.955±0.006	0.944	0.9546
CIFAR-10	supcon→res cat	0.9552±0.0023	0.9404	0.954

Table 5: Best residual construction per dataset (best base shown). Residual fine-tuning beats the discovery pipeline in 12/12 transfer cells. [†]aircraft prior champion carries a holdout-contamination caveat; the row here uses strict open-world parents. [‡]the one exp-70 record residuals do not beat (simclr-ft pre-discovery on texture).

5.2 Residual fine-tuning wins, and breaks the dissociation

The alternative construction: freeze the parent’s seen-class centroids, train a second head end-to-end on the residual $\mathbf{r} = \mathbf{z} - \mu_y$, and evaluate the concatenation $[\mathbf{z}; \mathbf{r}]$. The parent half keeps what supervision learned; the residual half is free to encode the within-class variation that supervision discarded — which is where fine-grained novelty lives.

Three things make this more than a leaderboard entry.

The winning residual objective splits by the same axis as the probe world. NPLM residual concatenation wins on fine-grained data (Cars 3/3, Flowers 2/3); plain softmax residual concatenation wins on texture and coarse data (DTD 3/3, Galaxy10 3/3). The estimator split from §4 reappears *inside* the residual construction, which we read as evidence that it is a property of what the objectives encode rather than of how they are deployed.

NPLM residual concatenations keep calibration while raising the probe. Cars/VISReg at probe 0.855 with per-event 0.263 and the best eucl and mahaT on Cars to date is **the first space in the campaign to hold both currencies at once**. Plain residual concatenations, by contrast, often buy their probe *with* calibration — Galaxy10/LeJEPa reaches probe 0.975 at per-event 0.001, which is a warning label, not a win.

The effect is essentially deterministic. Multi-seed checks give Cars/VISReg paired $\Delta = +0.148 \pm 0.004$ (3/3 seeds) and CIFAR-10 paired $\Delta = +0.0157 \pm 0.0004$ (5/5 seeds). We cite records with uncertainties throughout (0.833 ± 0.017 on Cars; 0.955 ± 0.006 on CIFAR-100) because several single-seed peaks in the campaign were fair-but-high draws, and one objective ranking (plain-res vs res-nplm on CIFAR-100) *flipped* under averaging.

Stacking is purity-gated. Running discovery on top of the residual winners stacks only where the residual space already separates the novelty: Flowers +0.021 (new record 0.906) and DTD/VISReg +0.015 (new record 0.862, finally beating simclr-ft), flat or slightly negative elsewhere. On fine-grained data the discovery fine-tune erodes exactly the residual concatenation’s probe directions.

6 Scale-free statistics see what distances miss

Every calibration statistic in §3 is a distance. Distances fail in a specific and predictable way: when novel classes sit *on* the data manifold, near some seen centroid, minimum-distance scores are blind by construction. This is the fine-grained regime — exactly the regime where pooling fails (§5).

We therefore add a *scale-free* statistic: the Levina–Bickel estimator of local intrinsic dimension,

$$\text{LID}_k(\mathbf{x}) = - \left[\frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{r_j(\mathbf{x})}{r_k(\mathbf{x})} \right]^{-1}, \quad (2)$$

computed against seen-class training references ($k = 20$, max 4000 refs). It depends only on *ratios* of neighbor distances, so it is invariant to the scale deformations that break both fixed- σ kernels and distance thresholds. Higher LID indicates novelty: holdout LID exceeds seen LID in nearly every space we measured — novelty lives in locally higher-dimensional neighborhoods.

space (CIFAR-100, 100-D)	agree@1	agree@5	dendro purity	silhouette
supcon (parent)	0.700	0.468	0.591	0.197
supcon_sigreg	0.690	0.460	0.603	0.199
supcon→res-nplm (concat)	0.670	0.452	0.564	0.188
supcon→res (concat)	0.640	0.406	0.514	0.162
simclr	0.570	0.366	0.457	0.118
nplm_bilinear	0.530	0.334	0.417	0.059
supcon→res (residual half)	0.490	0.288	0.365	0.029
lejepa	0.450	0.342	0.427	0.046

Table 6: Semantic structure against the CIFAR-100 coarse-label partition (chance agree@1 = 0.040). Semantics track *supervision*, not probe quality and not calibration quality.

Result On Flowers-102, LID reaches $\text{AUC} = 0.951\text{--}0.955$ using *no novelty labels at all*, exceeding the best supervised probe on that dataset (0.906). This is the only unsupervised score to beat a trained probe anywhere in the campaign, and it is a new Flowers novelty record.

Controls We treat a result of this shape with suspicion and ran the corresponding controls. *Holdout rotation*: over 10 random 10-class holdouts on a frozen trunk, Flowers LID gives 0.879 ± 0.018 (min 0.844) versus a $k\text{NN}$ -distance baseline at 0.692 ± 0.047 — classes 92–101 are not special. *Distance baselines*: LID beats $k\text{NN}$ -distance by $+0.08\text{--}0.12$ and euc1 by $+0.03\text{--}0.06$ everywhere on Flowers, with $\text{Spearman}(\text{LID}, k\text{NN-}d) = 0.55\text{--}0.92$ — correlated, but carrying independent signal. *Robustness*: bootstrap 95% CIs ± 0.006 ; per-holdout-class AUC minimum 0.86–0.92 (no single-class driver); k -sensitivity plateaus at $k \in [20, 50]$ and degrades to ≈ 0.78 at $k = 5$ (the ratio estimator needs neighbors). A rank ensemble with euc1 adds nothing — LID subsumes it.

And where it fails The same protocol on Cars gives 0.545 ± 0.040 , and on VISReg fine-tuned Cars spaces the ordering flips: $k\text{NN}$ -distance and euc1 beat LID. Aircraft and Galaxy10 sit at 0.56–0.66. LID’s advantage is regime-specific, and the regime is a dataset property, not a protocol artifact — the identical rotation protocol produces 0.88 on Flowers and 0.55 on Cars. The reading: Flowers holdouts sit on the flower manifold with small distances and mediocre distance-based scores, but a distinct local dimensional signature; Cars holdouts do not have one.

LID as a pool scorer Because LID is scale-free it should also fix the failure mode where a discovery fine-tune inflates the space and the distance-based pool threshold decays. It does: substituting LID for min-anchor-distance as the pool scorer holds round-2 purity at 0.60–0.68 where the distance scorer collapses to 0.18–0.33, a $\sim 3\times$ improvement, and *strictly dominates* as a pooling mechanism. But the downstream battery does not move (probe deltas $+0.000, -0.029, +0.007$). Above the purity gate, purity is not the binding constraint — the fine-tune dynamics are. This is a clean example of why a battery, rather than a single number, is necessary: a mechanism can be strictly better and operationally irrelevant at the same time, and only a multi-currency report shows both facts.

7 Interpretability, measured

A discovery that cannot be explained is not yet a discovery. We treat semantic legibility as a measurable property and find it dissociates from the other currencies as sharply as they dissociate from each other.

The founding question Are otters near beavers? On a CIFAR-100 supervised parent space with beaver held out entirely — never labeled, never in the corpus — the nearest-centroid table reads $\text{otter} \rightarrow \text{seal}(0.02)$, $\text{beaver}^*(0.08)$, $\text{bear}(0.11)$ and $\text{beaver}^* \rightarrow \text{otter}(0.08)$, $\text{shrew}(0.09)$, $\text{seal}(0.10)$, $\text{porcupine}(0.13)$. The never-labeled holdout lands mutually top-2 with otter purely from fine-tuning on the other 99 classes. The same holds on Galaxy10: the never-labeled edge-on bulge sits 0.06 from edge-on no bulge, its true sibling, with *cigar smooth* next — which is what an edge-on disk looks like. Open-world novelty settles where it semantically belongs, and this is the property that makes a discovered cluster explainable.

Semantics track supervision, not the probe The extreme case is Galaxy10 on DINO: `supcon-ft` recovers the four-group morphology partition *perfectly* (dendrogram purity 1.000, `agree@1` 0.9), while the label-free `nplm-bil-ft` — despite its calibration strengths — has essentially no semantic structure (purity 0.49, silhouette -0.08). The probe-versus-calibration dissociation is *also* a semantic-structure dissociation, and this is the strongest argument in the paper for reporting Currency IV separately: a practitioner selecting on calibration alone can land on a space in which no discovered cluster can be described.

Concatenation is semantically free Residual concatenations inherit the parent’s semantics exactly (Galaxy10 `res-nplm concat` purity 1.000, equal to parent) while adding the calibration half. On CIFAR-100 the record residual concatenation *tightens* the semantic ring around the held-out beaver (0.06–0.10) at near-parent purity. Meanwhile the residual half *alone* scrambles semantics by design (beaver $\rightarrow$ mushroom/crab/lobster, purity 0.365) — semantics ride in the parent half, and the concatenation restores them while keeping the probe gain. This is a strong practical argument for concatenation over replacement.

What “fine-grained mistakes” actually are On Cars, the space organizes by body style as much as by make (`make-agree@1` 0.40 against chance 0.04), with errors like Audi convertible $\rightarrow$ BMW convertible. These are not failures; they are visually coherent groupings that a make-based partition scores as wrong. We note this because superclass agreement metrics can penalize exactly the structure a scientist would want.

7.1 How spaces relate to each other

Cross-space measurements supply mechanisms for several claims above.

Supervision erases the pretraining base; self-supervision preserves it. The same supervised arm fine-tuned from two *different* bases reaches $\text{CKA} \approx 0.74$ on average — as similar as two *different supervised losses* on the same base. The label signal pulls every pretraining toward one shared supervised geometry, which is precisely why the regime rules of §4.1 are base-invariant. Label-free `simclr-ft` is far less base-invariant (CKA 0.46–0.62). Consistently, on DTD — where `simclr` wins the probe — `simclr-ft`’s *local* structure transfers across bases better than the supervised arms’ (mutual- k NN 0.28 vs 0.16): its texture neighborhoods are the transferable part.

NPLM spaces are reorganizations, not new information. The NPLM arms are geometric outliers everywhere (CKA 0.2–0.36, mutual- k NN 0.03–0.06 against every other arm, LLE weight-transfer ratio > 1) — yet they are highly *decodable from* softmax spaces (asymmetric ridge R^2 up to 0.97 at CKA 0.36). Calibration is a re-encoding of information the supervised space already contains. This explains why concatenation works so well and why a single loss struggles: the two currencies are different *codes* for overlapping content.

Residual children are mostly linear re-maps — and drift is capacity, not help. Parent-vs-child ridge R^2 is 0.97–0.98 on coarse data (the child is nearly a linear re-map; calibration is re-weighting) and 0.91–0.95 with mutual- k NN ≈ 0.2 on fine-grained data (locally reorganized, 5–10% genuinely new variance). Within the transfer grid the concatenation gains track the unexplained fraction. But CIFAR refutes the naive version of this rule: there children drift *much* further (R^2 down to 0.32) while producing the campaign’s *smallest* probe gains (+0.006–0.016 vs Cars’ +0.148). Unexplained variance is necessary and not sufficient — what pays is whether the new variance encodes the holdout.

Loss families have intrinsic-dimension fingerprints. TwoNN intrinsic dimension orders the arms identically in every cell: NPLM supervised most compressed ($\text{ID} \approx 2\text{--}3$), SIGReg/SSL arms mid (5–7), softmax and frozen highest (9–13). The Gaussian target and the density-ratio collapse are directly visible as dimension, which makes ID a cheap objective-provenance diagnostic.

8 A reporting protocol

We distill the campaign into a protocol we would want applied to any embedding space advertised for scientific discovery.

1. **Report all four currencies.** Probe-only or geometry-only conclusions are, in our experience, wrong roughly as often as they are right, because the currencies are anti-correlated by construction.
2. **State the number of seen classes, the embedding dimension, and the centroid-to-anchor distance.** These determine which corner of the design cube is even available (classwise marginals require $\dim \geq n_{\text{classes}}$).
3. **Never claim a probe gap below 0.02 from one seed.** Single-seed spread is ≈ 0.017 . Two rankings in this campaign flipped under averaging.
4. **Use annealed median-heuristic bandwidths** for any kernel two-sample statistic compared across spaces with different scales. A fixed bandwidth reports zero power on a space that is fine.
5. **Report pool purity next to any discovery gain.** A gain at purity < 0.15 is refinement, not discovery, and behaves differently (it depends on substrate alignment, not on the pool).
6. **Include at least one scale-free statistic.** Distance scores are structurally blind to on-manifold novelty; LID costs nothing and, in one regime, beat every supervised alternative we had.
7. **Audit semantics explicitly.** Superclass agreement, dendrogram purity, and a nearest-centroid table a domain expert can read. Calibration does not imply legibility — we have a space with strong calibration and silhouette -0.08 .

And a default recipe For a practitioner who wants one starting point: fine-tune end-to-end with supcon (adding a global SIGReg marginal if the data are fine-grained and low-shot), then train a residual head on $\mathbf{z} - \mu_y$ with an NPLM objective on fine-grained data or a softmax objective on coarse/texture data, and evaluate the concatenation. Add a discovery pass only if measured pool purity exceeds ≈ 0.3 . Score with the four-currency battery, including LID.

9 Limitations

Our evidence is entirely in the image domain; the regime rules are stated in terms of class counts and samples per class and we expect them to transfer, but we have not tested them on the scientific modalities (event-level collider data, spectra, time series) that motivate the work. The “discovery” setting is simulated by holding out known classes, which is a proxy: real novelty is not a withheld member of a curated taxonomy, and the Cars result — where novel classes are not geometrically outlying — suggests the proxy’s difficulty varies more than the datasets’ nominal difficulty does. Several conclusions rest on 3–5 seeds, which resolves 0.02-scale effects but not smaller ones. Our discovery loop is one specific algorithm (quantile pool, BIC k -selection, anchor merge, prototype fine-tune) and the purity-gating conclusion may be specific to it. Finally, the interpretability metrics require a superclass partition, which is available in our datasets and frequently is not in the settings we care about; the nearest-centroid tables remain available but require expert reading rather than a number.

10 Conclusion

Embedding spaces built for scientific discovery need to be evaluated as instruments, not as classifiers. When we do so — across 79 controlled experiments, seven datasets, three pretraining bases and ten objectives — the picture is consistent: linear readability and calibrated geometry are anti-correlated properties that no single loss in our design space achieves together; the repair is architectural (residual decomposition and concatenation) rather than a better loss; scale-free statistics reach novelty that distance-based scores cannot see, in a regime one can identify in advance; and semantic legibility is a third, separately measurable axis that tracks supervision alone. The distributional-prior program that LeJEPa and SIGReg opened up is, we think, the right foundation — specifying the marginal is what makes any of these quantities well posed. What it needs alongside it is a metrology: a standard battery, honest uncertainties, and the discipline to report the currency a space *lacks*.

Reproducibility statement

All experiments are scripted (`experiments/NN_*.py`, each with a mandatory `--quick smoke` mode) over a shared library (`supersig/`: losses, training loops, metrics, discovery). Result arrays are archived as `logs/expNN/*.npz` with fixed key conventions, condensed into per-dataset master tables under `docs/`. Seeds are fixed and identical across paired cells; the probe uses seeds $1000+s$. Known reproduction jitter from cuDNN nondeterminism is $\approx \pm 0.003$ in AUC and is stated wherever single-seed records are cited.

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